CS-471: Machine Learning

Week 5

Instructor: Dr. Hashir Kiani

Classroom Etiquette and Participation Guidelines

- Treat everyone with respect and courtesy at all times
- Speak only when it is your turn and raise your hand before contributing to the discussion
- Do not interrupt others while they are speaking
- Join the class on time
- Keep mobile phones on silent during the lecture

Lecture Outline

- Recap of Previous Lecture
- Practical Aspects of Machine Learning
- Decision Trees

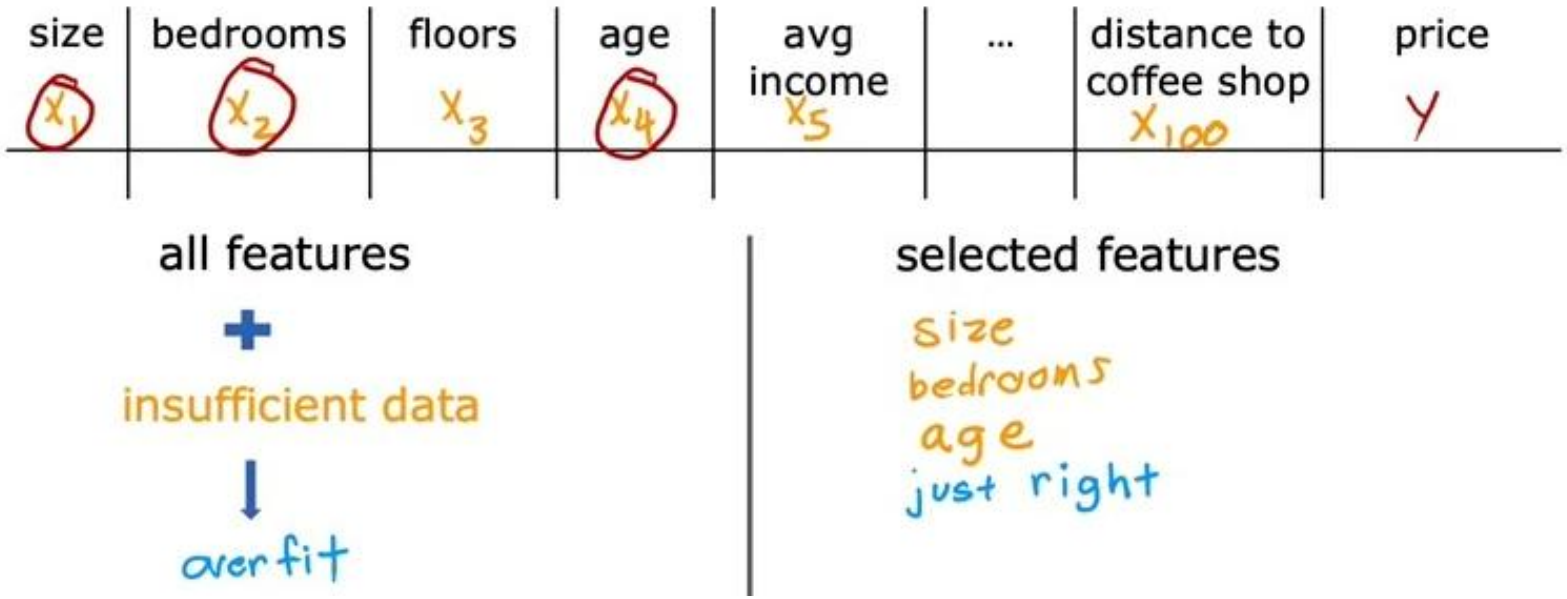
Recap of Previous Lecture

- How can we prevent overfitting?
- How can we prevent underfitting?
- What is vanilla cross validation?
- What is k-fold cross validation?
- Does using more data help with underfitting? Overfitting? How?

Cross-Validation

- Vanilla cross validation
- K fold cross validation for small datasets
- Leave one out cross validation when very small dataset
- Can be used to avoid overfitting

Feature Selection



- Can use fewer features.
- Choosing the most appropriate set of features is called feature selection
- Disadvantage of feature selection is that by using only a subset of the features, the algorithm is throwing away some of the information.

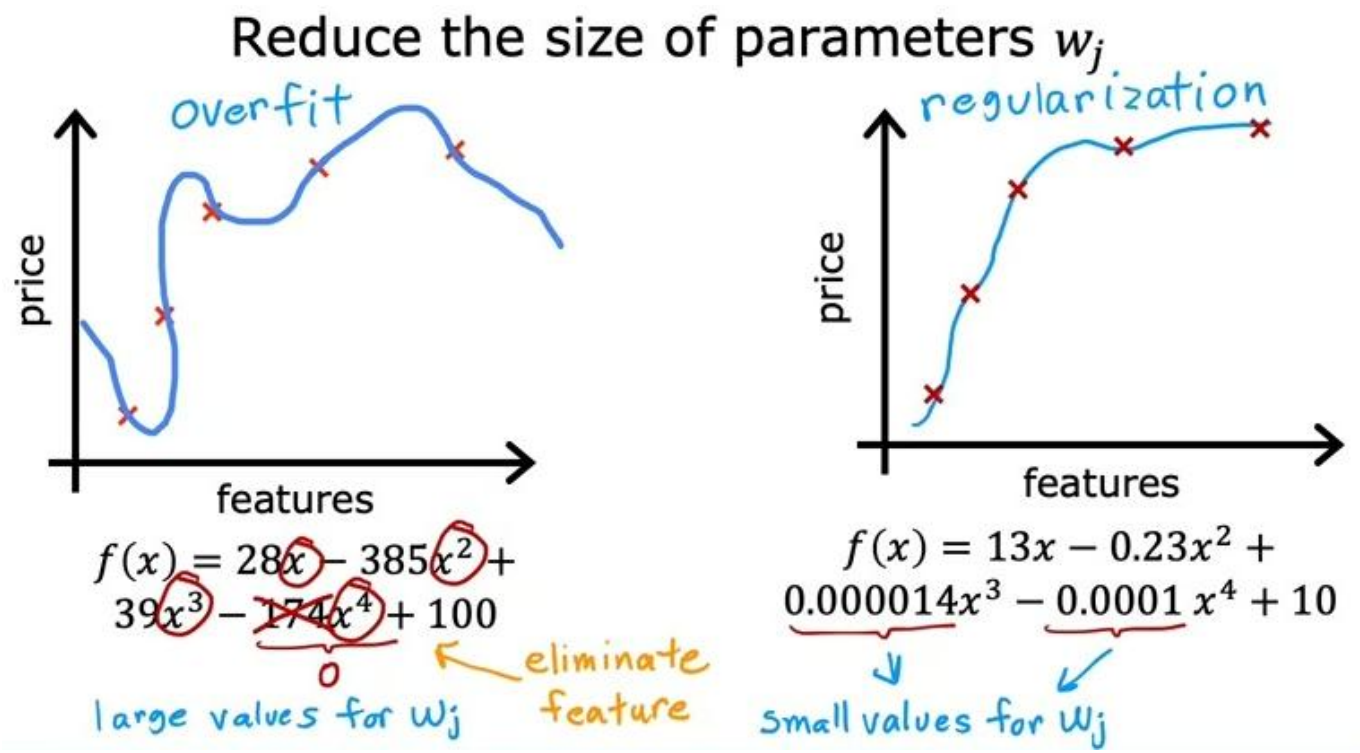
Feature Engineering

- Few features present
- Model is underfitting
- Feature Engineering is adding new features by using domain knowledge. These features better describe the relationship between input and output thus increasing model performance.

Regularization

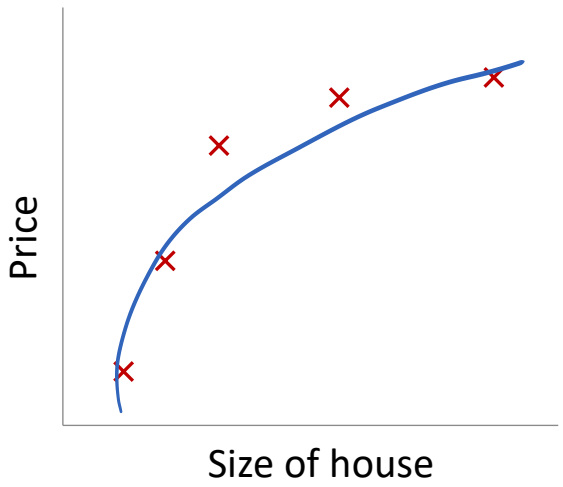
- Adds a penalty term to the loss function
- Penalizes large parameter values
- Two common types are L1 and L2 regularization
- L1 adds penalty term proportional to absolute value of model weights
- L2 adds penalty term proportional to square of model weights

Regularization

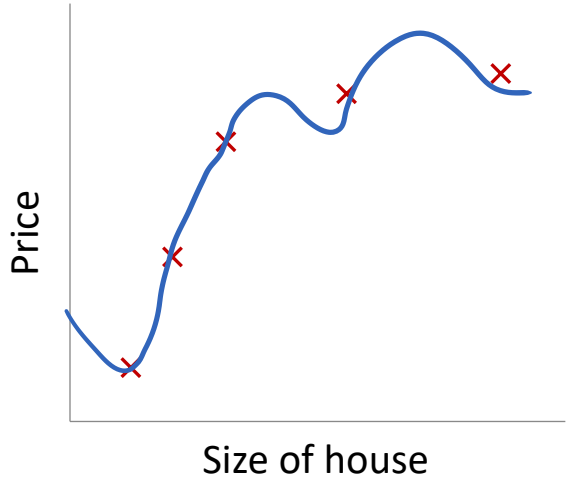


- Way to gently reduce the impacts of selected features instead of elimination.
- Allows you to keep all of your features, but they just prevents the features from having an overly large effect, which can sometimes cause overfitting.

Regularization Intuition



$$\theta_0 + \theta_1 x + \theta_2 x^2$$



$$\theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4$$

Suppose we penalize and make θ_3, θ_4 really small.

$$\min_\theta \frac{1}{2m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)})^2 \quad + 1000\,\theta_3^2 + 1000\,\theta_4^2$$

Regularization

- Small values for parameters
  - "Simpler" hypothesis
  - Less prone to overfitting

- L2 Regularization:

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)})^2 + \frac{\lambda}{2m} \sum_{j=1}^{n} \theta_j^2$$

Regularization

- Balances two goals: fitting the data well and preventing overfitting.
- First term: Minimizes the difference between predictions and actual values.
- Second term: Keeps model parameters small to reduce overfitting.
- Lambda (λ): Controls the trade-off between these two goals. A larger λ reduces overfitting, while a smaller λ fits the training data more closely.

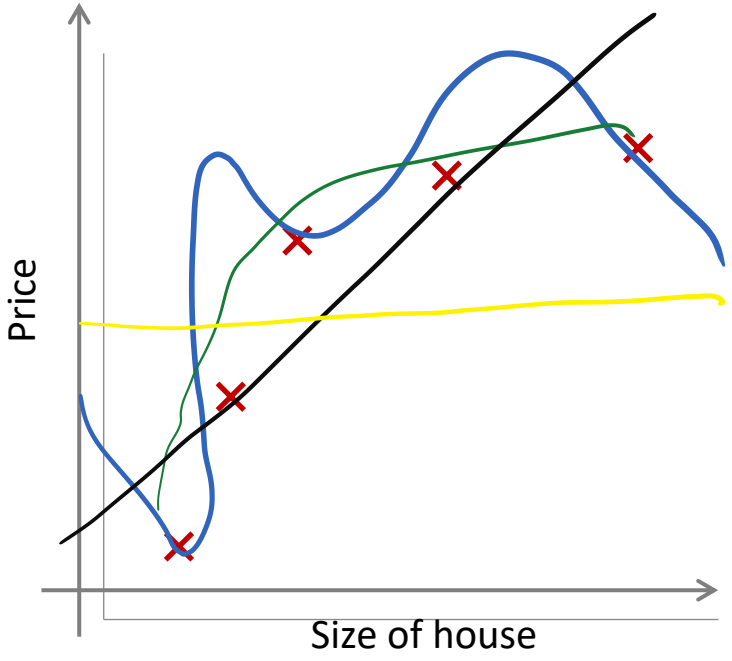
Regularization

- If lambda is 0 the model will overfit.
- If lambda is enormous like 10^3, this model will underfit.
- You want some value of lambda that is in between that more appropriately balances these first and second terms of trading off, minimizing the mean squared error and keeping the parameters small.

Regularization

$$J(\theta) = \frac{1}{2m}\left[\sum_{i=1}^{m}(h_\theta(x^{(i)}) - y^{(i)})^2 + \lambda\sum_{j=1}^{n}\theta_j^2\right]$$

$$\min_\theta J(\theta)$$



$\lambda = 0$

$\lambda = 5$

$\lambda = 10$

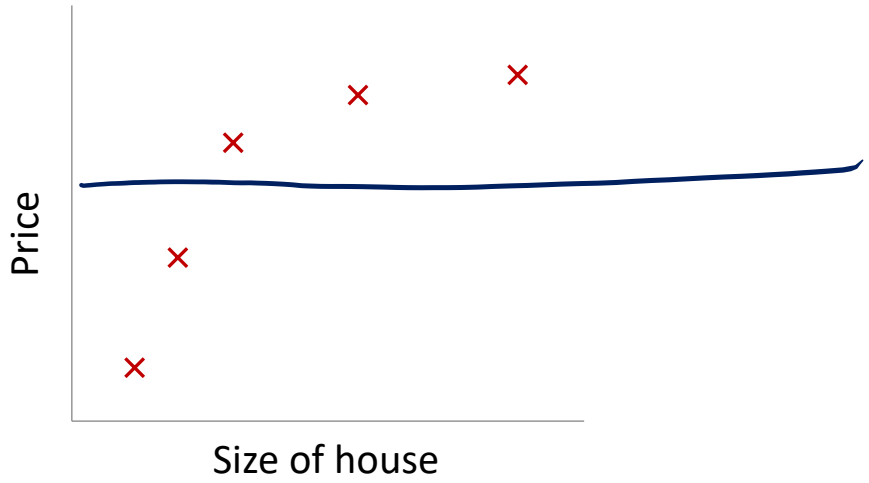
$\lambda = 100$

Regularization

In regularized linear regression, we choose θ to minimize

$$J(\theta) = \frac{1}{2m}\left[\sum_{i=1}^{m}(h_\theta(x^{(i)}) - y^{(i)})^2 + \lambda \sum_{j=1}^{n} \theta_j^2\right]$$

What if λ is set to an extremely large value (perhaps for too large for our problem, say $\lambda = 10^{10}$)?



$$\theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4$$

Gradient Descent with Regularized Linear Regression

Repeat $\{$

$$\theta_0 := \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)}) x_0^{(i)}$$

$$\theta_j := \theta_j - \alpha \left[\frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)}) x_j^{(i)} + \frac{\lambda}{m} \theta_j \right]$$

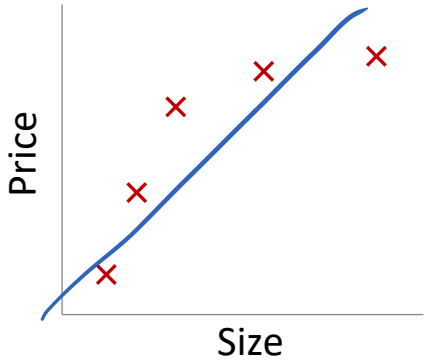
$$(j = \cancel{0}, 1, 2, 3, \ldots, n)$$

$\}$

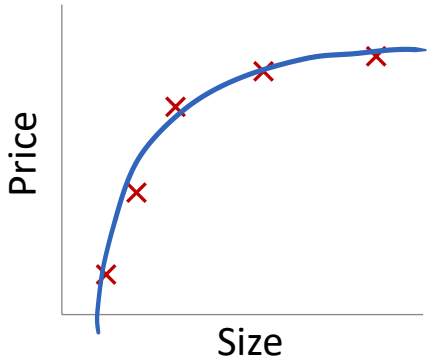
Regularized Linear Regression

Model: $h_\theta(x) = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4$

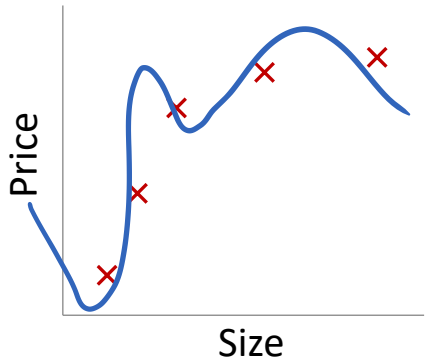
$$J(\theta) = \frac{1}{2m}\sum_{i=1}^{m}(h_\theta(x^{(i)}) - y^{(i)})^2 + \frac{\lambda}{2m}\sum_{j=1}^{m}\theta_j^2$$



Large λ
High bias (underfit)
$\lambda = 10000.\ \theta_1 \approx 0, \theta_2 \approx 0, \ldots$
$h_\theta(x) \approx \theta_0$



Intermediate λ
"Just right"



Small λ
High variance (overfit)

Choosing Regularization Parameter

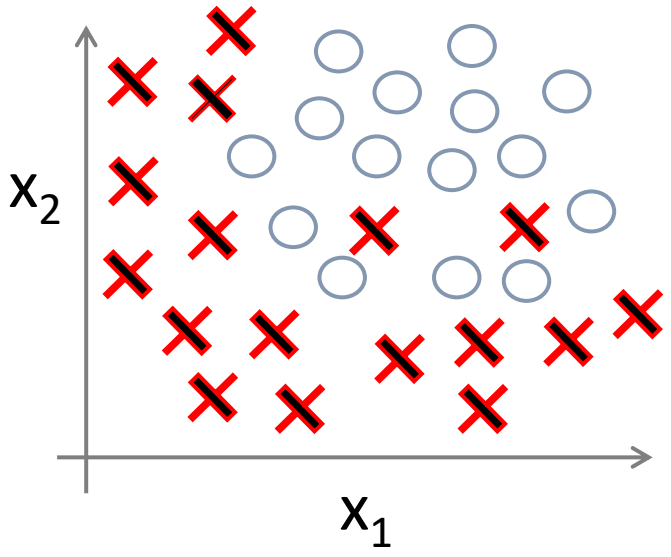
Model: $h_\theta(x) = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4$

$$J(\theta) = \frac{1}{2m}\sum_{i=1}^{m}(h_\theta(x^{(i)}) - y^{(i)})^2 + \frac{\lambda}{2m}\sum_{j=1}^{m}\theta_j^2$$

1. Try $\lambda = 0$
2. Try $\lambda = 0.01$
3. Try $\lambda = 0.02$
4. Try: $\lambda = 0.04$
5. Try $\lambda = 0.08$

12. Try $\lambda = 10$

Regularized Logistic Regression



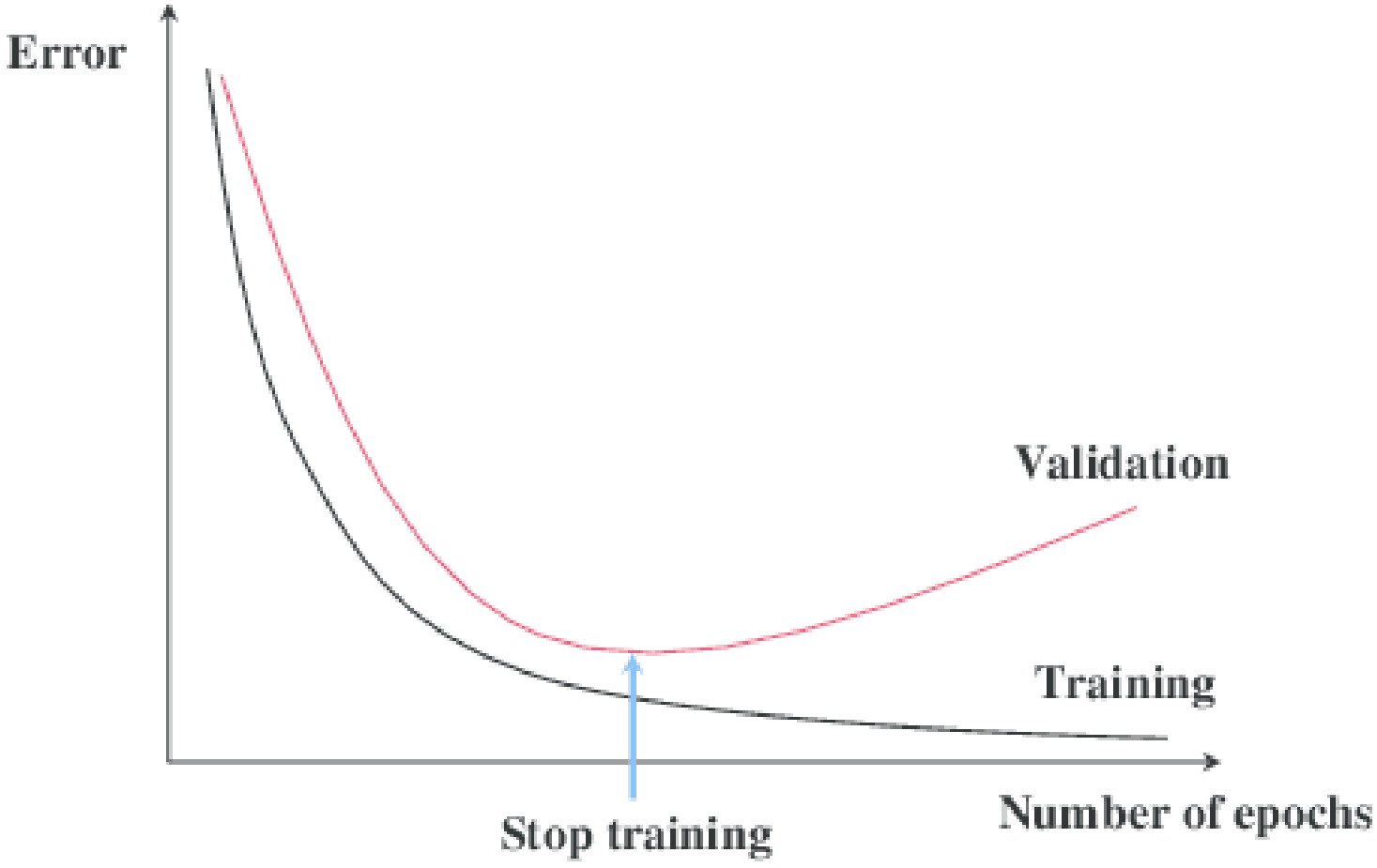
$$h_\theta(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_1^2 + \theta_3 x_1^2 x_2 + \theta_4 x_1^2 x_2^2 + \theta_5 x_1^2 x_2^3 + \dots)$$

Cost function:

$$J(\theta) = -\left[\frac{1}{m}\sum_{i=1}^{m} y^{(i)} \log h_\theta(x^{(i)} + (1 - y^{(i)}) \log (1 - h_\theta(x^{(i)}))\right] + \frac{\lambda}{2m}\sum_{j=1}^{n} \theta_j^2$$

Early Stopping

- Stop before the network starts to over fit



Decision Trees

Training Set

Example	Input Attributes				Output Plays Golf
	Outlook	Temperature	Humidity	Windy	
$\boldsymbol{x}_1$	Rainy	Hot	High	False	$y_1 = \text{No}$
$\boldsymbol{x}_2$	Rainy	Hot	High	True	$y_2 = \text{No}$
$\boldsymbol{x}_3$	Overcast	Hot	High	False	$y_3 = \text{Yes}$
$\boldsymbol{x}_4$	Sunny	Mild	High	False	$y_4 = \text{Yes}$
$\boldsymbol{x}_5$	Sunny	Cool	Normal	False	$y_5 = \text{Yes}$
$\boldsymbol{x}_6$	Sunny	Cool	Normal	True	$y_6 = \text{No}$
$\boldsymbol{x}_7$	Overcast	Cool	Normal	True	$y_7 = \text{Yes}$
$\boldsymbol{x}_8$	Rainy	Mild	High	False	$y_8 = \text{No}$
$\boldsymbol{x}_9$	Sunny	Mild	Normal	False	$y_9 = \text{Yes}$
$\boldsymbol{x}_{10}$	Rainy	Mild	Normal	True	$y_{10} = \text{Yes}$
$\boldsymbol{x}_{11}$	Overcast	Mild	High	True	$y_{11} = \text{Yes}$
$\boldsymbol{x}_{12}$	Sunny	Mild	High	True	$y_{12} = \text{No}$

Validation Set

Example	Input Attributes				Output Plays Golf
	Outlook	Temperature	Humidity	Windy	
$\boldsymbol{x}_1$	Rainy	Cool	Normal	False	$y_1 = \text{Yes}$
$\boldsymbol{x}_2$	Overcast	Hot	High	True	$y_2 = \text{Yes}$

Decision Trees

- Can be used for both numeric and non-numeric data
- Simple and Interpretable
- Useful when some characteristics might not matter
- Useful when some characteristics might matter only under certain conditions
- Build a tree of tests for different attributes
- Could have multiple trees
- Best tree is one with minimum cost and minimum levels
- We want the smallest possible tree

Questions?